

Intermediate (make search conditional) — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint has 2 depth levels: **Basic** (the plain idea) and **Intermediate** (proper Python, plus how a real router handles the messy edge cases). Read Basic first even if you already know Python — it's the fastest way to spot exactly what Intermediate adds.

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Hint 1 — The idea, and the exact pieces

Basic Version

Right now, your graph either always searches or leaves it up to the model to decide inside one call. This exercise wants a separate decision step, made by the graph itself, before the model ever gets to reason: "does this specific task need the document search, or not?"

Think of it as a fork in the road that happens right after the task comes in, not something buried inside the model's own reasoning.

Things to use:

- A plain function that looks at the task and returns a short label, like `"search"` or `"skip"`.
- `graph.add_conditional_edges(from_node, routing_function, {"search": "search_node_name", "skip": "reason_node_name"})`.
- Two test questions: one that clearly needs your documents, one that clearly doesn't (like "hello, how are you?").

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Intermediate Version

This is Doc09's `add_conditional_edges` mechanism, applied to a real decision: "route to search, or skip it."

A conditional edge needs a plain Python function that looks at the current state and returns the *name* of the next node to go to — it doesn't do any work itself, it just decides:

```
# search_tool_integration_practice.py – Intermediate section
def should_search(state: dict) -> str:
    ...
    return "search" # or "skip"
```

Then you register it:

```
# search_tool_integration_practice.py – Intermediate section
graph.add_conditional_edges(
    "router", should_search, {"search": "search_node", "skip": "reason_node"},
)
```

The exact pieces:

- **The routing function's return value** — it must exactly match one of the keys in the mapping dict you pass to `add_conditional_edges`. A typo here fails silently in confusing ways (LangGraph raises an error about an unknown node, which can look unrelated to the real cause).
- **Where the decision lives** — a plain keyword/heuristic check is a fine first version; a second, smaller model call classifying "needs search: yes/no" is the more solid real-world version, at the cost of one extra call.
- **The two mapped branches** — one path goes to your search node, the other skips straight to your reasoning node, both eventually rejoining before the final answer.

A keyword list and a single model call both share the same weakness: they're each wrong in a different, predictable direction. A keyword list misses real search-needing questions phrased without its exact words ("what's our time-off rule?" instead of "policy"). A model call catches those, but costs one extra round trip on *every single task*, even the 90% that are obviously one way or the other — and it isn't perfectly deterministic either, so the same question can route differently on different runs.

The real design question isn't "keywords or a model call" — it's "what's the cheapest check that's right most of the time, and when is it worth paying for a more expensive, more accurate check

instead?"

A hybrid answers that directly: run the cheap keyword check first. If it's confident (a clear keyword hit, or a clearly conversational greeting with no keywords and no question mark), trust it and skip the model call entirely. Only fall back to the model classification call for the genuinely ambiguous middle — task text with no keyword hit that still isn't obviously a greeting.

The extra pieces needed for a hybrid router:

- A "confident skip" check — something unmistakably conversational, like a short message with no question mark and no keyword hit, routes straight to `"skip"` without ever calling the model.
- A "confident search" check — a clear keyword hit routes straight to `"search"`, same reasoning.
- Only the leftover, ambiguous cases fall through to the one-line model classification call from Hint 2.
- A cost/accuracy note worth writing down: this hybrid still isn't perfect (there's no free lunch here) — it just spends the expensive check only on the tasks where the cheap one is least trustworthy.

Sketch which cases you'd route with keywords alone, and which you'd fall through to the model, before Hint 2.

Difference between Basic and Intermediate: Basic names the conditional-edge mechanism and the two branch labels. Intermediate shows the real `should_search` signature and wiring, names the keyword-vs-model tradeoff, and treats it as a spectrum instead of a binary choice — cheap and fast for the clear cases, falling through to the more expensive, more accurate check only for the genuinely ambiguous ones — which is the difference between a router that's "good enough for most" and one that's deliberately built to spend its accuracy budget where it actually matters.

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Hint 2 — The plan, and almost the whole thing

Basic Version

```
function should_search(state):
    look at the task text
    if it contains words like "document", "policy", "according to"
    (or similar clues):
        return "search"
    otherwise:
        return "skip"

wire it up:
    add_conditional_edges(from the router node, should_search,
        {"search": search_node, "skip": reason_node})

test:
    run with a question mentioning your documents -> should route to search
    run with "hello, how are you?" -> should skip straight to reasoning
```

Here's almost the whole thing — just try running it and reading it line by line:

```
# search_tool_integration_practice.py – Intermediate section
def should_search(state):
    task = state["messages"][-1].content.lower()
    if "document" in task or "policy" in task:
        return "search"
    return "skip"

graph.add_conditional_edges(
    "router", should_search, {"search": "search_node", "skip": "reason_node"},
)
```

Expected output if you run just this: nothing on its own — invoke the graph with a document-mentioning question and a plain greeting, and check which node each one lands on.

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Intermediate Version

```
def define:
    def should_search(state: dict) -> str:
        task = state["messages"][-1].content.lower()
        keywords = ["document", "policy", "according to", "the file says"]
        if any(keyword in task for keyword in keywords):
            return "search"
        return "skip"

    register:
        graph.add_conditional_edges(
            "router",
            should_search,
            {"search": "search_node", "skip": "reason_node"},
        )

    test:
        invoke with a document-referencing question -> confirm it went
            through search_node
        invoke with an unrelated greeting -> confirm it skipped straight
            to reason_node
```

```
# search_tool_integration_practice.py – Intermediate section
def should_search(state: dict) -> str:
    task = state["messages"][-1].content.lower()
    keywords = ["document", "policy", "according to"]
    has_match = False
    for keyword in keywords:
        if keyword in task:
            has_match = True
    if has_match:
        return "search"
    return "skip"
```

A keyword check is a reasonable first version but is easy to fool. Write the `add_conditional_edges` wiring and both test invocations yourself, then compare against the [Solution](#).

Once that's working, add the confident short-circuits and a model-call fallback for the ambiguous middle:

```
function should_search(state):
    task = the task text
    if task has a keyword: return "search"    # confident hit
    if task has no keyword AND looks conversational (short, no "?"):
        return "skip"    # confident skip
    otherwise (ambiguous):
        ask a small model call: "does this need document search? yes/no"
        return "search" if yes else "skip"
```

Here's almost the whole thing — fill in the missing piece yourself:

```
# search_tool_integration_practice.py – Intermediate section
from langchain_core.messages import HumanMessage

KEYWORDS = ["document", "policy", "according to", "the file says"]

def should_search(state: dict) -> str:
    task = state["messages"][-1].content
    lowered = task.lower()
    has_keyword = False
    for keyword in KEYWORDS:
        if keyword in lowered:
            has_keyword = True

    if has_keyword:
        return "search"

    looks_conversational = len(task.split()) <= 6 and "?" not in task
    if looks_conversational:
        return "skip"

    # your turn: the ambiguous middle case -- fall through to a
    # one-line model classification call, the same pattern as
    # Hint 1's note, and return "search" or "skip" based on its answer
    ...
```

Fill in the fallback classification call yourself, then compare all of your finished versions against the [Solution](#).

Difference between Basic and Intermediate: Basic's pseudocode and near-complete code always return based on one keyword check, no matter how ambiguous the task. Intermediate adds two

confident short-circuits (a clear keyword hit, a clearly conversational message) and only falls through to a model call for whatever's left over — the same idea Hint 1 describes, made concrete: spend the expensive check only where the cheap one is genuinely unsure.

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Full solution: [Show me the solution](#)